



Unit 3 · Outcome 2 · Sustainable Development

Circular economy thinking & the tools that test a decision

KK05 VCE Environmental Science · Units 3 & 4 · Exam study summary

Digital Vector EnviroSci · single-topic mastery summary with worked good-vs-bad answers, exam traps, command words and practice.



Every management strategy is a bet

00 The decision problem Every management strategy is a bet A good environmental decision is rarely obvious. Strategies cost money, carry risks, and help some stakeholders while hurting others. Across this chapter you've met the sustainability principles (KK03) and the challenges that strain them (KK04). But principles alone don't tell you which strategy to fund. Two managers can both "support intergenerational equity" and still disagree about whether to build a fishway or buy back water licences. That's where assessment tools come in: structured ways to

From take–make–waste to make–use–return

01 Circular economy thinking From take–make–waste to make–use–return A circular economy is a system designed so that resources are kept in use for as long as possible and "waste" is treated as a resource in the wrong place. Most of our economy is linear : we take raw materials, make products, use them briefly, then dispose of them. The line has two open ends — a tap draining finite resources at one end and a bin overflowing with waste and pollution at the other. A circular economy bends that line into a loop: outputs from one process become inputs for an

The word examiners are really testing: integrated

02 What "integrated" means The word examiners are really testing: integrated "Integrated sustainability assessment" means weighing a decision across all three dimensions of sustainability at once — ecological, economic and sociocultural — rather than one in isolation. From KK02 you know sustainable development sits at the overlap of three dimensions. An integrated assessment refuses to look at just one. A purely economic analysis might green-light draining the wetland for irrigation because the dollar return is highest. An integrated analysis forces the

Ranking what could go wrong

03 Tool 1 · Qualitative risk analysis Ranking what could go wrong Risk analysis asks two questions about each thing that could go wrong: how likely is it? and how bad would it be? The combination gives the level of risk. It's qualitative because it uses descriptive categories — words like "possible" and "major" — rather than precise numbers. That suits environmental problems where exact probabilities are unknown but expert judgement is available. The two ingredients: Likelihood How probable the event is — rated rare → unlikely → possible → likely → almos

Do the benefits outweigh the costs?

04 Tool 2 · Cost–benefit analysis Do the benefits outweigh the costs? Cost–benefit analysis (CBA) lists everything a strategy will cost and everything it will deliver, values them on a common scale where possible, and checks whether total benefits exceed total costs. Define the strategy (and the alternatives you're comparing it against). List all costs and benefits across the ecological, economic and sociocultural dimensions — including externalities (impacts on third parties) and intangibles. Assign values , converting to dollars where you can and ackno

Two tools, two jobs

05 Choosing the tool Two tools, two jobs They answer different questions, so the best decisions often use both. Qualitative risk analysis Cost–benefit analysis Asks: what could go wrong, and how badly? Asks: is it worth it overall? Output: a risk rating (low → extreme) per hazard Output: a net benefit (benefits – costs) Best when uncertainty & potential harm are the worry Best when comparing whether a strategy pays off Mostly descriptive (words, categories) Mostly quantitative (values, dollars) where possible For Boorla, a manager might first risk-analys

Decode the verb before you write

◆ COMMAND WORDS

06 Command words & traps Decode the verb before you write Most marks lost here aren't knowledge — they're the wrong kind of answer for the command word. Identify / State Name it, no explanation. "A cost is the irrigation water foregone." Describe Say what it is and its features. Outline how the matrix combines likelihood and consequence. Explain Give the how / why . Why does circular thinking reduce waste? Because by-products become inputs. Apply Use the tool on the case study — actual Boorla risks, costs and benefits, not generic ones. Analyse Break the

Build a full-mark sentence with Point · Explain · Link

07 P·E·L writing trainer Build a full-mark sentence with Point · Explain · Link Extended-response marks come from structure. Draft each part, then reveal a model to compare. Train it P · E · L Question (3 marks): A manager proposes delivering an environmental flow to the Boorla Wetlands. Explain how cost–benefit analysis could be used to assess this strategy as part of an integrated sustainability assessment. Point State what the tool does Explain How it works, across dimensions Link Tie back to the decision/case Reveal model answer Clear Model — full ma

Five graded questions · 19 marks

★ PRACTICE & SELF-MARK

08 Exam practice Five graded questions · 19 marks Attempt each in your logbook first. Then reveal the weak vs strong responses, check the marking guide, and self-mark — your score feeds the dock and the rail. Q1 Distinguish 2 marks Distinguish between a linear economy and a circular economy. Reveal responses ✗ Weak response · 0–1 / 2 A linear economy is normal and a circular economy recycles stuff so it's better for the environment. [Vague; "recycles stuff" only captures part of circularity, and there's no clear contrast of the two systems.] ✓ Strong res

★ Good vs bad worked answers

Study the contrast — the same question answered two ways. The strong column shows the moves that earn the marks.



The P-E-L answer structure — make a Point, support with Evidence, then Link to the question.

✗ WEAK ANSWER

A linear economy is normal and a circular economy recycles stuff so it's better for the environment. [Vague; "recycles stuff" only captures part of circularity, and there's no clear contrast of the two systems.]

✓ STRONG / MODEL ANSWER

A linear economy follows a one-way "take–make–dispose" path , drawing finite resources and discarding waste. A circular economy keeps resources in use in a loop — designing out waste and returning outputs as inputs so the system regenerates rather than depletes.

✗ WEAK ANSWER

They could recycle their water and not waste it. Recycling is good for the wetland because it's more sustainable. [Names one idea but doesn't explain the mechanism, only references one principle loosely, and ignores the case detail.]

✓ STRONG / MODEL ANSWER

Circular thinking would design waste out by capturing nutrient-rich farm run-off in constructed reed beds that strip nutrients before water re-enters the river, preventing algal blooms in the wetland. It would keep water in use by treating Connawarra's wastewater for reuse in irrigation, so less is drawn fresh from the dam. Treated biosolids could be returned to farmland as fertiliser , closing a nutrient loop. Together these regenerate the wetland by improving the quality and reliability of downstream flows.

✗ WEAK ANSWER

This is a high risk because killing fish is really bad. They should be careful. [Gives consequence but no likelihood, no rating logic, and a vague mitigation.]

✓ STRONG / MODEL ANSWER

The likelihood of a blackwater event is "possible" (it depends on temperature and flood size), and the consequence is "major" (a fish kill harms a Ramsar ecosystem and native species). Combining these on a risk matrix gives a "high" risk rating . The risk could be reduced by delivering the flood in cooler months and releasing it gradually , lowering the likelihood that oxygen is stripped from the water — moving the rating down the matrix.

✗ WEAK ANSWER

The cost is the money to build the fishway and the benefit is more fish. So they should build it because fish are good. [Only the economic cost and one ecological benefit; no sociocultural dimension, no integration, weak judgement.]

✓ STRONG / MODEL ANSWER

Economic: costs include construction and the irrigation water foregone (an opportunity cost), while benefits include tourism and jobs from a healthier wetland. Ecological: costs include a short-term blackwater risk ; benefits include recovered native fish and birds and water-filtering ecosystem services . Sociocultural: a key benefit is cultural reconnection for the Boorla people as flooding returns to significant sites. Because the integrated benefits across all three dimensions outweigh the costs, the analysis supports proceeding — though some benefits (cultural value) are hard to price and may be undervalued.

✗ WEAK ANSWER

CBA is useful because it shows costs and benefits. It is good and helps make decisions. [Only strengths, repeated; no limitation, no criteria, no conclusion — not an evaluation.]

✓ STRONG / MODEL ANSWER

Strength: CBA is useful because it makes trade-offs explicit and comparable , supporting transparent, defensible decisions and allowing options to be ranked. Limitation: its usefulness is reduced when intangible and non-market values — biodiversity, cultural significance, future generations' interests — cannot be reliably priced , so they risk being undervalued or omitted, biasing the result toward easily-costed economic outcomes. Judgement: CBA is therefore most useful as part of an integrated assessment , used alongside risk analysis and stakeholder input, rather than as a stand-alone dollar test.